

Reasoning Meets Evidence: LLMs for Interpretable Insomnia Detection with Evidence Extraction in Clinical Notes

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Background & Motivation

Insomnia is often **under documented** in clinical notes described implicitly through symptoms like *"unable to sleep"*, *"restless"*, or *"fatigued"* rather than explicit diagnoses.

Automated detection from EHRs must go beyond classification: it requires **clinically grounded reasoning** and **faithful evidence localization** at the character level.

Shared Task Overview

SUBTASK 1 Binary Classification

Predict whether a clinical note indicates insomnia ("yes"/"no"). Evaluated using F1-score on the positive class.

SUBTASK 2 Multi-label + Evidence Extraction

Predict four rule labels (Definition 1, Definition 2, Rule B, Rule C) and extract supporting character-level spans. Evaluated with micro-F1 + exact/partial match.

Dataset Challenges

- **Implicit symptoms** - insomnia inferred from fatigue, agitation, restlessness
- **Medication ambiguity** - benzodiazepines used for multiple conditions
- **Label imbalance** across four rule categories
- **Clinical heterogeneity** - abbreviations, fragmented sentences, long notes

Methodology

We explore two complementary paradigms across **four** systems:

Parameter-efficient Fine-tuning SUBTASK 1

- **Gemma3 1B — QLoRA**
- 4-bit NF4 quantization · LoRA rank 8, alpha 16 · instruction-style prompts · 8 epochs · LR 2×10^{-4} · cosine scheduler · paged_adamw_32bit
- **Qwen3 1.7B — LoRA**
- Low-rank adapters on q/k/v/o/gate/up/down projections · 12 epochs · LR 1×10^{-4} · Unsloth + SFTTrainer · sequence length 1024

Few-shot Prompting SUBTASK 2

- **Qwen3 8B — Few-shot v1**
- Structured examples for joint rule prediction + span extraction. Output as JSON with character offsets. Focused on classification recall.
- **Qwen3-8B — Few-shot v2**
- Stricter span alignment instructions · minimal, text-faithful extraction · diverse example set · improved exact-match at slight classification cost.

Error Analysis

- ❑ **Implicit symptoms** - "very tired" not linked to insomnia without explicit cues.
- ❑ **Medication ambiguity** - *Lorazepam* flagged as insomnia indicator even for other indications.
- ❑ **Span boundary errors** - semantic identification correct but character offsets imprecise.
- ❑ **Negation** - "no sleep issues" misclassified in complex sentence structures.
- ❑ **Long-context** - evidence distributed across distant note sections.

Results

0.7333
Best F1
SUBTASK 1

0.6535
Micro-F1
SUBTASK 2

0.5192
Partial-F1
EVIDENCE

Subtask 1 — Binary Classification

Model	Precision	Recall	F1
Gemma-1B (QLoRA)	1.00	0.58	0.73
Qwen-1.7B (LoRA)	0.52	0.63	0.57

Subtask 2 — Multi-label Classification

Model	Precision	Recall	F1
Qwen-8B (FS v1)	0.58	0.75	0.65
Qwen-8B (FS v2)	0.54	0.68	0.60

Subtask 2 — Evidence Span Extraction

Model	Exact F1	Partial F1
Qwen-8B (FS v1)	0.14	0.36
Qwen-8B (FS v2)	0.40	0.52

Key Findings & Conclusion

- **Efficiency beats scale** - Lightweight QLoRA Gemma-1B outperforms larger models on binary classification (F1 0.73 vs 0.57).
- **Reasoning-localization trade-off** - v2 gains +0.16 exact-F1 on spans but loses 0.05 on classification F1.
- **Span extraction bottleneck** - Exact character-level matching (0.40) lags far behind partial matching (0.52), highlighting a key NLP gap.